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from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.neural_network import MLPClassifier
from sklearn.metrics import accuracy_score, log_loss

iris = load_iris()
X = iris.data
y = iris.target

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42)

scaler = StandardScaler()
X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)

models = [
    ("1 Hidden Layer (10)", (10,), "relu"),
    ("2 Hidden Layers (20,10)", (20,10), "relu"),
    ("2 Hidden Layers (20,10)", (20,10), "tanh"),
    ("3 Hidden Layers (30,20,10)", (30,20,10), "logistic")
]

for name, layers, activation in models:
    mlp = MLPClassifier(
        hidden_layer_sizes=layers,
        activation=activation,
        max_iter=1000,
        random_state=42
    )

    mlp.fit(X_train, y_train)

    y_pred = mlp.predict(X_test)
    y_prob = mlp.predict_proba(X_test)

    accuracy = accuracy_score(y_test, y_pred)
    loss = log_loss(y_test, y_prob)

    print("\nModel:", name)
    print("Activation Function:", activation)
    print("Accuracy:", round(accuracy, 4))
    print("Loss:", round(loss, 4))
```

```
Model: 1 Hidden Layer (10)
Activation Function: relu
Accuracy: 1.0
Loss: 0.059
```

```
Model: 2 Hidden Layers (20,10)
Activation Function: relu
Accuracy: 1.0
Loss: 0.0312
```

```
Model: 2 Hidden Layers (20,10)
Activation Function: tanh
Accuracy: 1.0
Loss: 0.0468
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```
Model: 3 Hidden Layers (30,20,10)
Activation Function: logistic
Accuracy: 1.0
Loss: 0.0698
```

```
/usr/local/lib/python3.12/dist-packages/sklearn/neural_network/_multilayer_percept
warnings.warn(
```